

ISILKUL, Russian Federation

WMO: 286880

Lat: 54.9374N

Lon: 71.2702E

Elev: 126

StdP: 99.82

Time Zone: 6.00 (E06)

Period: 94-18

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-33.9	-31.1	-36.9	0.1	-33.7	-34.1	0.2	-31.0	8.1	-10.0	7.3	-8.7	1.2	250	0.626

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.3	30.5	18.8	28.6	18.3	26.9	17.7	20.8	27.0	19.9	25.8	19.0	24.8	3.5	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.7	13.7	23.2	17.8	12.9	22.5	16.8	12.1	21.8	60.7	27.0	57.5	25.7	54.6	24.8	30.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8.4	7.4	6.4	-36.4	33.4	3.8	2.2	-39.1	35.0	-41.3	36.3	-43.5	37.5	-46.2	39.1
			-36.5	22.5	3.8	2.1	-39.2	24.0	-41.4	25.2	-43.5	26.4	-46.2	27.9
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	2.0	-17.7	-15.4	-7.3	4.2	12.5	17.8	18.6	16.8	10.7	3.6	-6.8	-13.8	(6)
(7)		DBStd	14.35	7.85	7.22	6.09	6.05	5.25	4.40	3.69	4.06	4.57	4.89	7.92	7.98	(7)
(8)		HDD10.0	3824	858	711	536	189	34	3	0	2	45	204	506	737	(8)
(9)		HDD18.3	6093	1116	944	794	424	190	61	43	79	234	457	756	996	(9)
(10)		CDD10.0	916	0	0	0	16	113	237	267	212	65	6	0	0	(10)
(11)		CDD18.3	143	0	0	0	0	10	46	52	31	4	0	0	0	(11)
(12)		CDH23.3	1515	0	0	0	6	161	487	508	301	52	0	0	0	(12)
(13)		CDH26.7	435	0	0	0	1	41	159	141	84	11	0	0	0	(13)
(14)	Wind	WSAvg	3.1	2.7	2.8	3.3	3.6	3.6	3.0	2.8	2.8	2.9	3.2	3.3	3.1	(14)
(15)	Precipitation	PrecAvg	368	16	13	17	23	29	49	65	51	27	28	28	21	(15)
(16)		PrecMax	561	37	27	52	62	62	97	110	120	59	73	65	42	(16)
(17)		PrecMin	243	2	1	0	4	3	14	11	19	8	4	6	5	(17)
(18)		PrecStd	78	10	6	12	14	15	18	26	27	15	16	14	11	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	0.3	1.2	7.4	23.7	30.6	32.3	33.0	31.9	28.0	18.6	7.5	1.6	(19)
(20)			MCWB	-0.3	0.2	4.1	12.7	16.8	18.7	20.0	19.0	15.2	10.8	5.4	0.2	(20)
(21)		2%	DB	-1.7	-0.7	3.6	19.6	27.4	30.5	29.9	28.9	24.3	16.1	5.2	0.3	(21)
(22)			MCWB	-2.5	-1.6	1.6	10.6	15.1	18.7	19.5	18.6	13.8	10.0	3.9	-0.5	(22)
(23)		5%	DB	-3.6	-2.4	2.2	16.7	24.9	28.4	27.9	26.6	21.0	13.3	3.7	-1.4	(23)
(24)			MCWB	-4.4	-3.2	0.7	9.8	14.1	18.3	19.2	17.9	13.2	8.5	2.4	-2.2	(24)
(25)		10%	DB	-5.9	-4.4	0.9	13.7	22.3	26.1	26.2	24.5	18.4	11.0	1.8	-3.4	(25)
(26)			MCWB	-6.5	-5.2	-0.3	8.3	13.1	17.3	18.8	17.3	12.1	7.3	0.7	-4.2	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-0.3	0.3	4.3	13.7	18.0	21.7	21.9	21.4	16.7	12.4	6.0	0.6	(27)
(28)			MCDWB	0.3	1.1	6.7	22.0	27.1	28.9	27.9	27.5	25.0	16.6	7.1	1.0	(28)
(29)		2%	WB	-2.4	-1.6	2.0	11.6	16.1	20.3	21.0	20.0	15.2	10.7	4.1	-0.5	(29)
(30)			MCDWB	-1.7	-0.8	3.2	18.0	24.7	27.0	27.0	25.7	21.7	14.6	5.2	0.3	(30)
(31)		5%	WB	-4.4	-3.1	0.9	10.0	14.9	19.1	20.2	19.0	14.0	9.0	2.4	-2.1	(31)
(32)			MCDWB	-3.7	-2.4	2.0	15.9	23.4	26.2	26.1	24.6	20.0	12.8	3.5	-1.4	(32)
(33)		10%	WB	-6.5	-5.1	-0.2	8.5	13.8	18.0	19.4	18.0	12.7	7.5	0.9	-4.2	(33)
(34)			MCDWB	-5.8	-4.4	0.9	13.5	21.1	24.5	24.9	23.0	17.6	10.7	1.7	-3.5	(34)
(35)	Mean Daily Temperature Range	MDBR	7.4	8.2	8.6	9.3	11.9	10.9	10.3	10.5	10.0	7.6	5.9	7.0	(35)	
(36)		5% DB	MCDBR	7.8	6.9	7.5	14.5	15.6	14.4	13.4	14.3	14.2	12.2	5.9	7.0	(36)
(37)			MCWBR	7.7	6.5	6.3	7.6	6.7	5.7	5.7	6.0	6.4	7.4	5.2	6.6	(37)
(38)		5% WB	MCDBR	7.8	6.7	6.9	13.4	14.0	12.2	11.3	11.5	12.8	11.0	5.6	6.9	(38)
(39)			MCWBR	7.8	6.4	6.0	7.6	7.0	5.4	5.7	5.8	6.3	7.1	5.2	6.7	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.284	0.316	0.373	0.411	0.404	0.407	0.432	0.432	0.373	0.354	0.320	0.287	(40)	
(41)		taud	2.050	2.019	1.943	2.112	2.228	2.298	2.218	2.185	2.313	2.238	2.104	2.011	(41)	
(42)		Ebn at Noon	613	731	769	805	838	840	807	778	782	691	558	502	(42)	
(43)		Edh at Noon	68	102	140	137	129	122	130	126	97	82	65	57	(43)	
(44)	All-Sky Solar Radiation	RadAvg	0.98	2.06	3.61	4.65	5.64	6.00	5.57	4.50	3.14	1.61	0.79	0.63	(44)	
(45)		RadStd	0.07	0.09	0.29	0.41	0.42	0.38	0.40	0.37	0.25	0.22	0.11	0.08	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (9 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page